



ME 340-2: Final Report

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Introduction and Overview of Part Design

Our group of four set out to design and manufacture a customizable Among Us character as part of our plastic injection molding project. Our goal was to create an engaging product that allows for interchangeable accessories. To achieve this, we designed the main character's body, along with two accessories:

- A mustache, which attaches to a hole in the character's face.
- A hat, which attaches to a pin on top of the character's head.

The project involved several key stages, including conceptualization, detailed CAD modeling, mold design, manufacturing, and final assembly, helping us gain hands-on experience with injection molding and repeatable manufacturing. This report will explore each phase in detail, covering the design process, mold fabrication, manufacturing execution, and final assembly, while also highlighting the challenges and constraints encountered throughout the injection molding workflow.

Concept Development:

Shown below are our team's initial sketches during the brainstorming phase, as well as the first iterations of our character and accessory designs. In Figure 1, our team's initial idea of our part can be seen in the form of a rough sketch.



Figure 1: Initial brainstorming phase Among Us character with a mustache and a hat

Further brainstorming led to the selection of a final design with specific attachment features, such as a hole in the face for the mustache and a pin on the head for the hat (shown in Figure 2). These shown dimensions were adjusted to ensure proper fit and manufacturability.

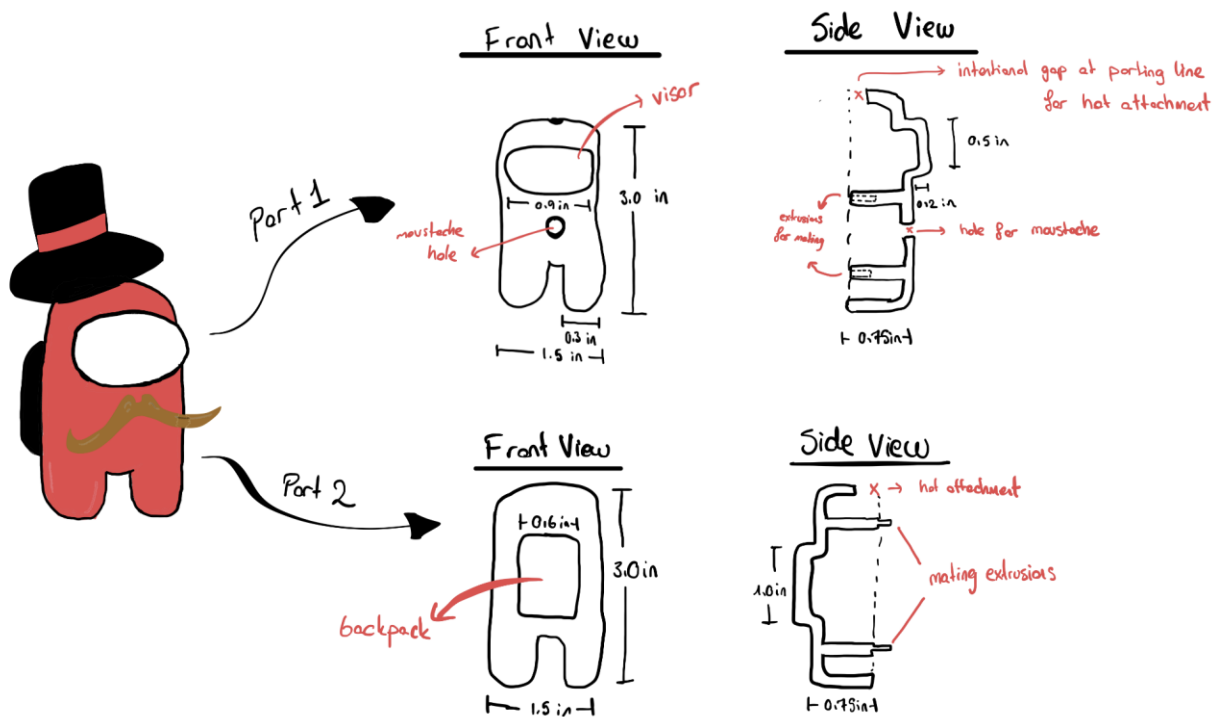


Figure 2: Body sketches with rough dimensions

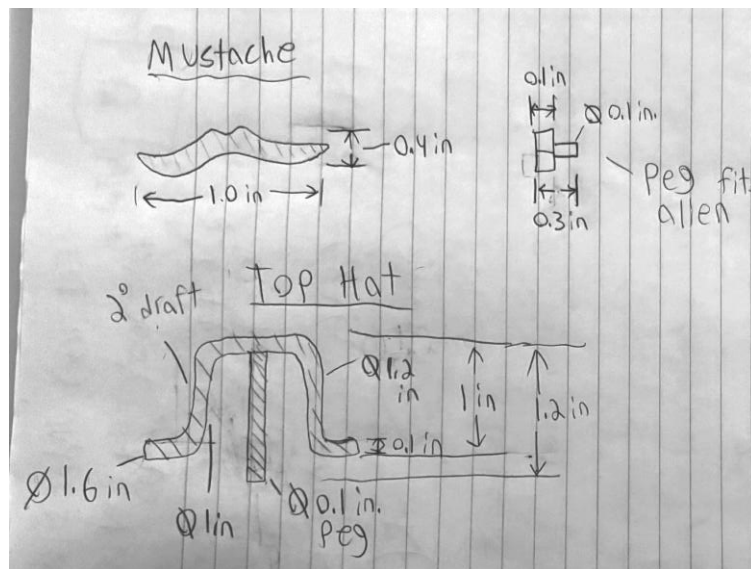


Figure 3: Mustache and hat attachments with rough dimensions

From these sketches, we transitioned into CAD modeling to refine our designs. Focusing on:

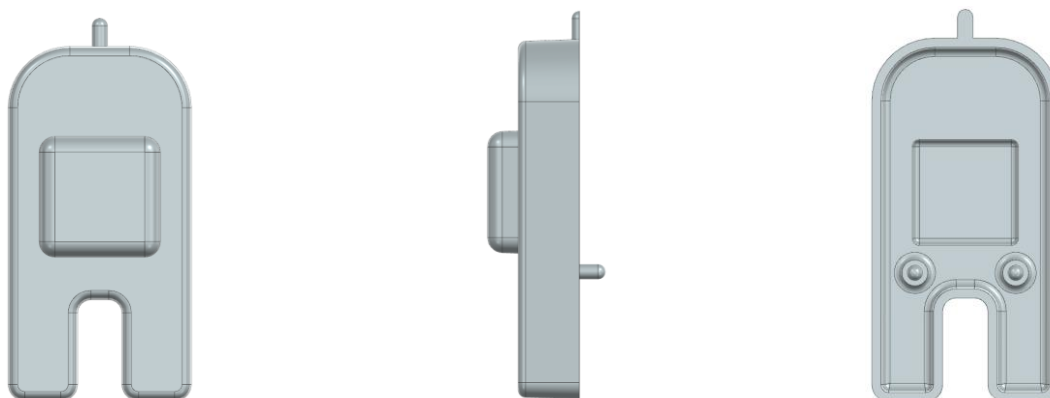
- Refine the design to ensure manufacturability.
- Maintaining uniform wall thickness to minimize shrinkage and warping.
- Adding draft angles to facilitate part ejection from the mold.
- Ensuring correct fits for the attachable accessories.

Finalized CAD Images:

1. Main Body Front



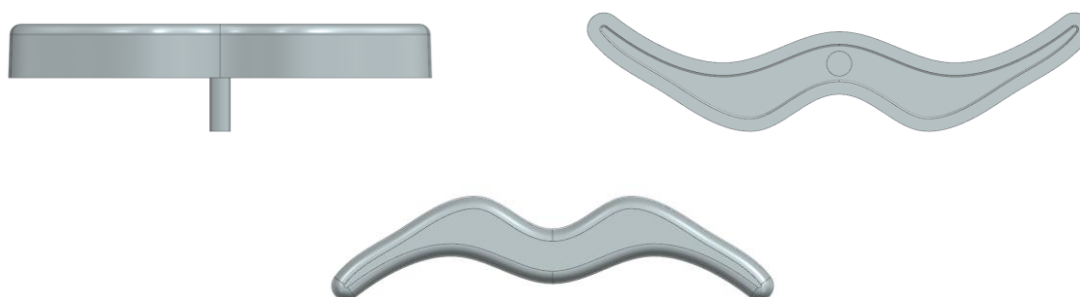
2. Main Body Back



3. Hat



4. Mustache



Part Design

When designing an injection-molded component, our team took into account several factors that must be considered to ensure manufacturability and part quality. One of the most important design principles is maintaining uniform wall thickness to minimize net shrinkage and warping. For our Among Us character,

we ensured that the body and accessory walls remained within the recommended range for polypropylene, as shown in Table 1. Maintaining a consistent wall thickness helps prevent defects such as sink marks and uneven cooling, which could affect the final assembly.

Section Size	Wall Thickness (inches)
Short Sections (< 0.5 in)	0.025
Small Sections (0.5 – 1 in)	0.035
Average Sections (1 – 8 in)	0.075
Large Sections (> 8 in)	0.125 – 0.185

Table 1: Recommended Wall Thickness for Polypropylene

To improve manufacturability, sharp corners were avoided, as they create stress concentrations and cooling inconsistencies. Instead, curved transitions were implemented where needed, with an inside radius of half the wall thickness and an outside radius of 1.5 times larger than the wall thickness. This helps distribute stress evenly and ensures smooth ejection from the mold.

All vertical walls have a **2-degree draft angle** on them, and as seen below in the part images, the edge that outlines the part has both an exterior and interior fillet. This makes it easier to remove the part from the mold after it has formed.

For assembly, we implemented a pin-hole interference fit system to allow the mustache and hat to securely attach while remaining removable. Instead of placing a hole on top of the head and a pin in the head, which would have separated the front and back halves of the Among Us character during assembly, we opted for a different approach. We designed a half-pin on the front half and a matching half-pin on the back half, with a hole in the hat that would accommodate both halves when assembled. This design not only secured the hat in place but also helped hold the two character halves together, ensuring a more structurally stable and seamless assembly. The hole for the mustache was similarly designed to provide a balanced fit, preventing excessive play while allowing for easy insertion and removal. The holes were strategically placed at the recommended 3 diameters apart from nearby walls to reduce stress in these areas, ensuring a strong and durable connection.

To create our hole features, we utilized the provided dowel pins. These pins were press-fit into our molds for use during injection and created holes of exactly 0.125” in diameter. Therefore we had to size our molded pins around this hole size. To create the pins, we drilled holes with a bit one size up from 0.125”. This was a #30 bit (diameter of 0.1285”). The calculation of design stress for interference fit is below:

$$E_h = 183 \text{ ksi} \quad \nu_h = 0.45 \quad E_s = 183 \text{ KSI} \quad \nu_s = 0.45 \quad d = 0.1285''$$

$$D = 0.25'' \quad d_1 = 0.125'' \quad I = d - d_1 = 0.0035'' \quad W = \frac{1 + (\frac{d}{D})^2}{1 - (\frac{d}{D})^2} = 1.718$$

$$I = (\frac{Sd * d}{W}) * ((\frac{W + \nu_h}{E_h}) + (\frac{1 - \nu_s}{E_s})) \quad 0.0035 = (\frac{Sd * 0.1285}{1.718}) * ((\frac{1.718 + 0.45}{183,000}) + (\frac{1 - 0.45}{183,000}))$$

$$Sd = 3.15 \text{ ksi}$$

This is roughly 60% of the yield stress of polypropylene, right in the range we want for a press-fit connection.

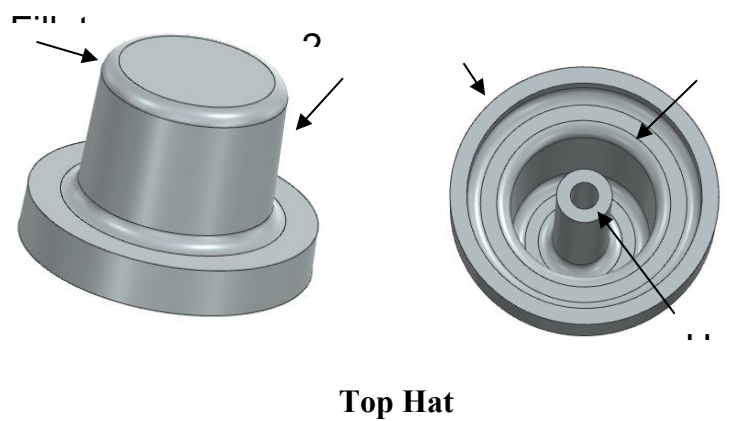
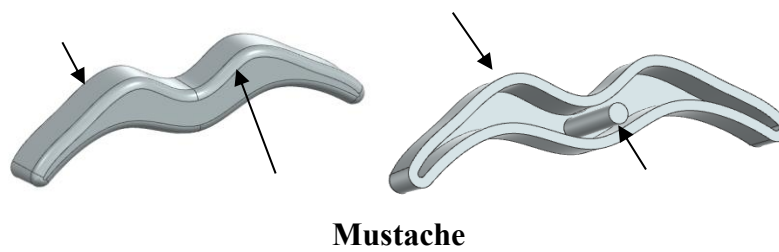
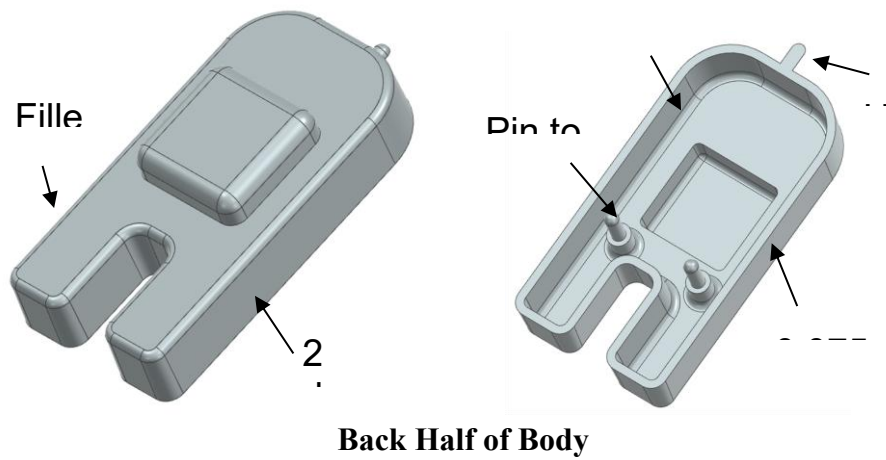
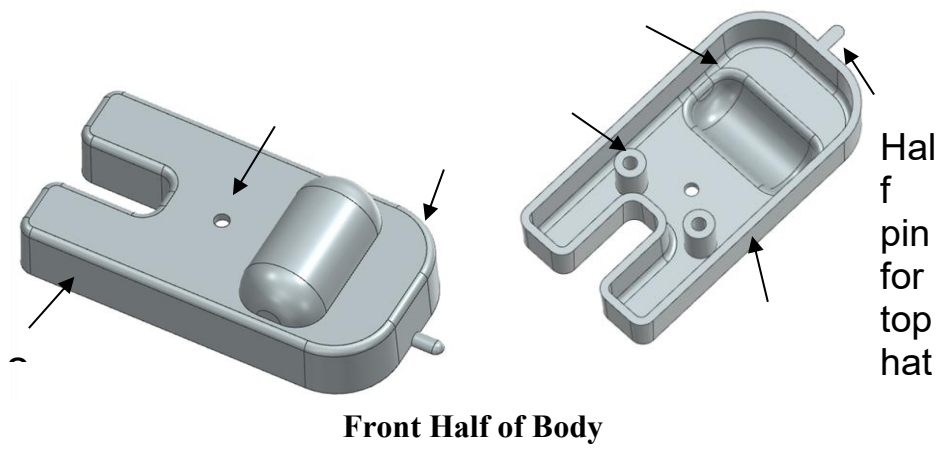
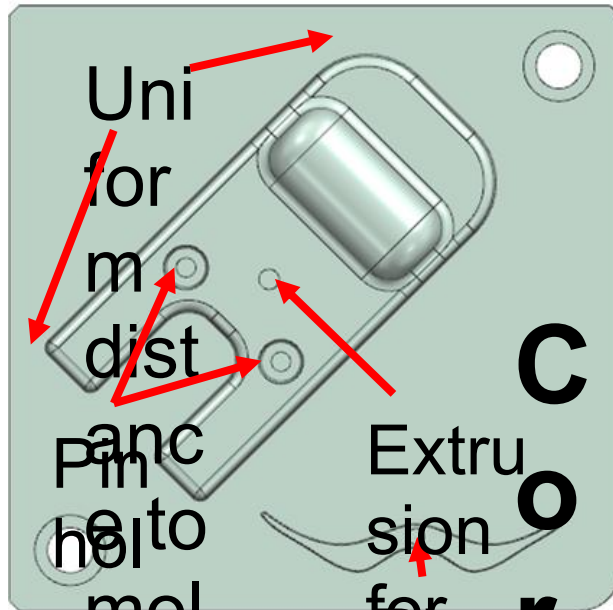


Figure 4: CAD images of the mold halves, highlighting design features

Review of Mold Design

Front of Body and Mustache Molds:

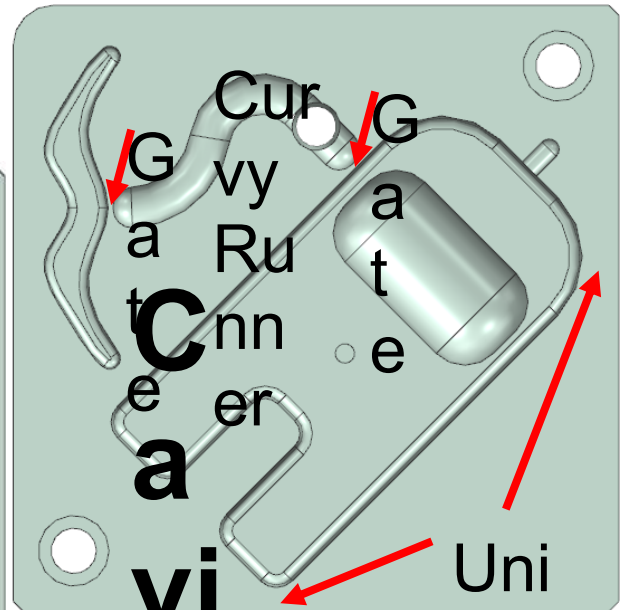
Core (Mold 1)



Uniform distance
Pin hole
Extrusion
Pin hole
Mustache Assembly

Core
Pin hole
Mustache Assembly

Cavity (Mold 2)



Cavity

Uniform Wall Distance

Back of Body and Hat Molds:

Core (Mold 3)

Cavity (Mold 4)

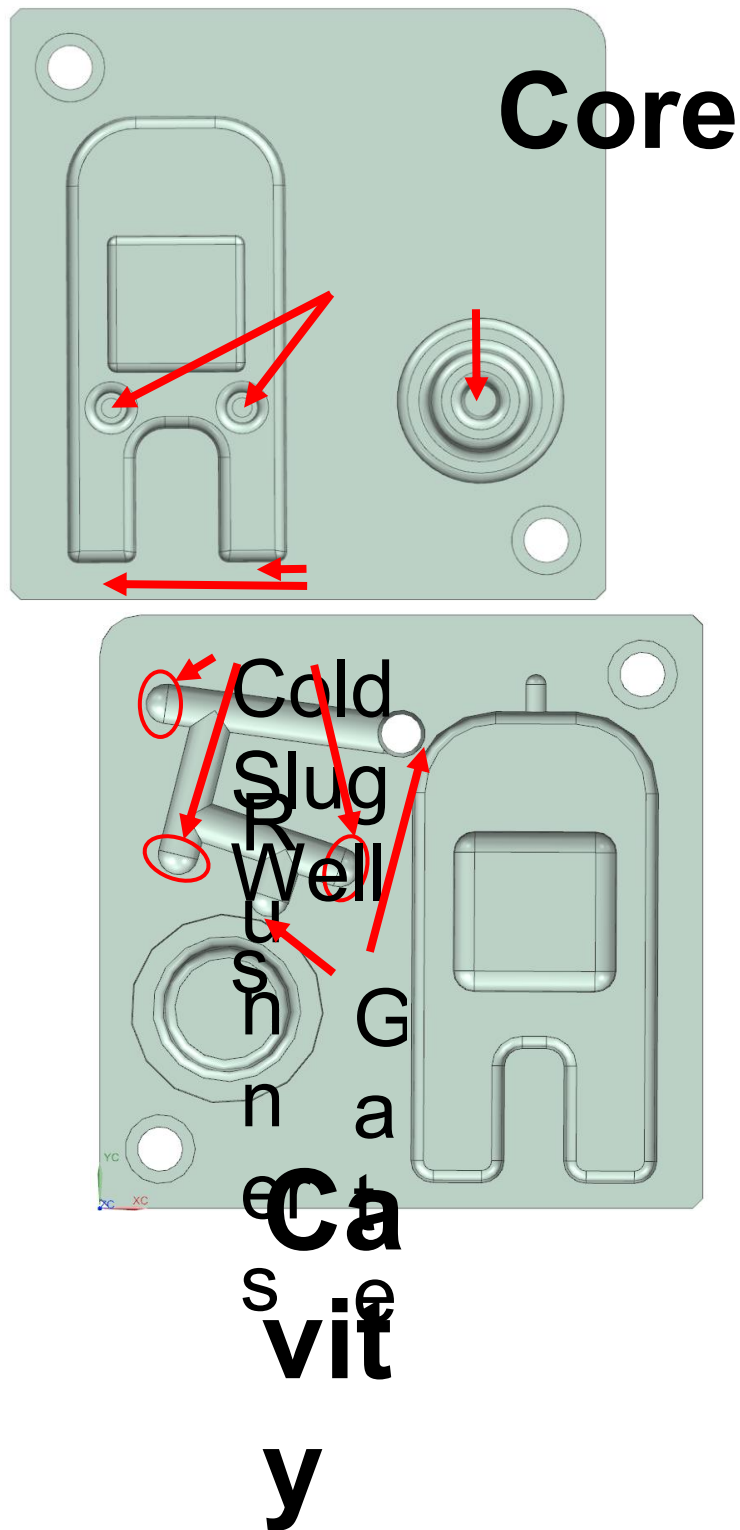


Figure 5: Cores and Cavities

Parting Line Selection: We chose the parting lines based on the geometry of the part to ensure ease of ejection, minimize the appearance of visible seams, and minimize any post-production finishing. As seen in Figure 6, the parting line allowed us to put assembly features (pin holes and extrusions) on the core and relatively simpler machining.

Part Placement: Part placement within the mold was optimized to facilitate uniform filling and cooling. We kept at least 0.2 inches between the part and the mold walls to prevent flash. As can be seen in Figure 5, the main bodies on the two mold pairs have different orientations due to the hat and mustache having different fill times. The part was placed 0.03” from the runners, which was to enable us to make the gate with files.

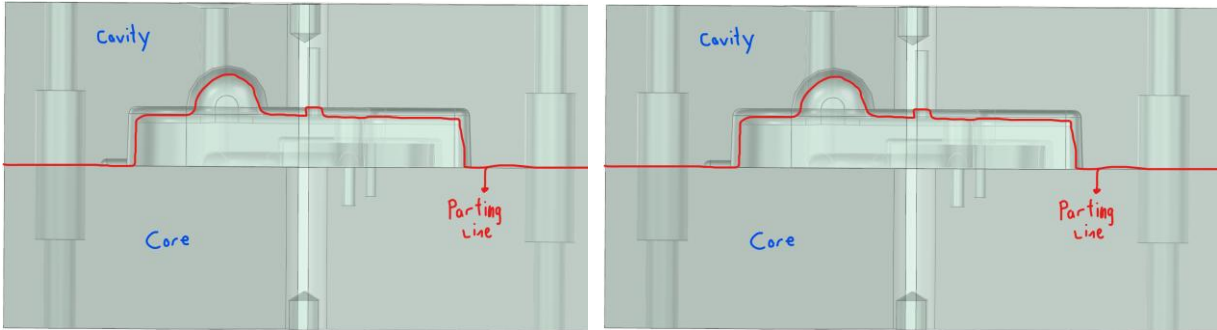


Figure 6: Main Body Parting Liner in Molds 1&2 and 3&4 respectively

Shrink Compensation: We rounded all the sharp corners and ensured uniform part thickness to prevent warping and dimensional instability that results from shrinkage.

Runner Design: The runner system was designed to ensure balanced material flow to all cavities. From Figure 7, it can be seen that the runner on the left is shorter and simpler. This is because the mustache had a smaller volume than the hat. The runner that goes to the hat, on the right in Figure 7, features cold slug wells used to remove the cold front of plastic travel and also delay the time the flow reaches the hat to ensure uniform filling.

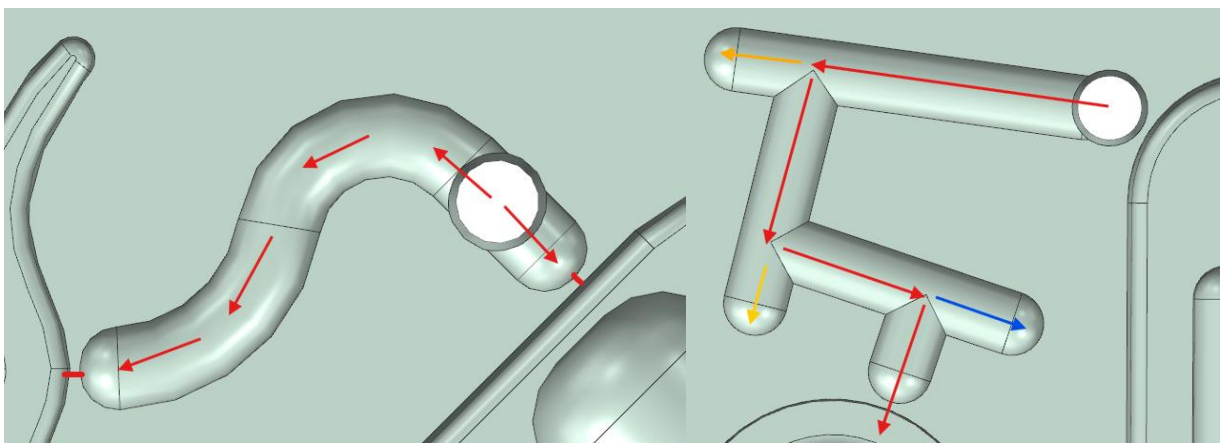
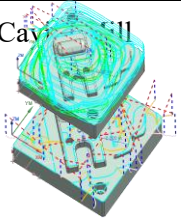
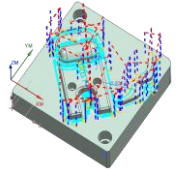
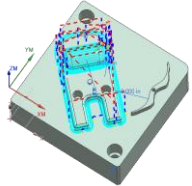
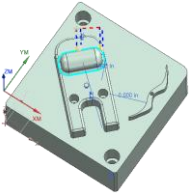
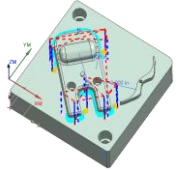
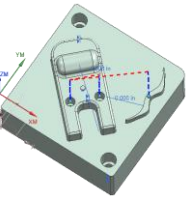


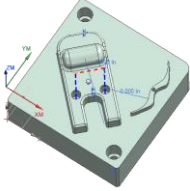
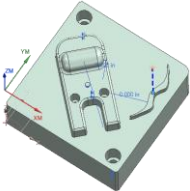
Figure 7: Runner Design and Flow Map

Review of Mold Manufacturing (CAM)

The following section outlines the CAM processes and operations used to manufacture each mold part.

Table 2: Core mold for the Front and the Mustache CAM Processes & Operations

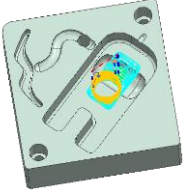
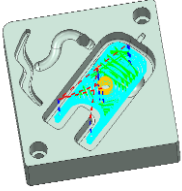
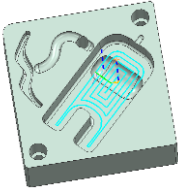
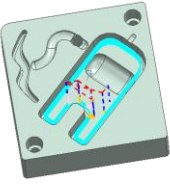
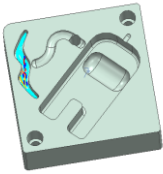
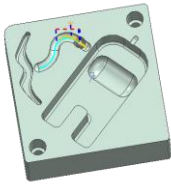
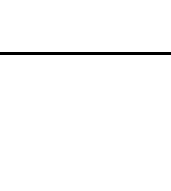
Process	Parameters	Tool	Discussion
3D Adaptive Milling	Mill Rough	$\frac{3}{4}$ in Flat End Mill	Remove material
	Mill Semi-Rough	$\frac{3}{4}$ in Flat End Mill	Remove material left by the Adaptive milling and form the Among Us Front and Mustache
	Mill Semi-Rough	$\frac{1}{4}$ in Flat End Mill	Remove material in areas that the $\frac{3}{4}$ in Flat End Mill could not get into
	Mill-Finish	$\frac{1}{4}$ in Ball End Mill	Finish operation to create a good surface finish on the Among Us Front and make the top fillets on the Among Us Front
	Mill-Finish	$\frac{1}{8}$ in Ball End Mill	Mill out the fillets on the bubble part of the Among Us Front
	Mill-Finish	$\frac{1}{8}$ in Tapered Flat End Mill	Create the draft around the Among Us Front and inside the stepped pin
	Center Drilling	Center Drill	Center drill on areas that will be drilled out in the following operations

Drilling 	Hole-making	#31 Drill Bit	Create the holes for the pins in the Among Us Front
Drilling 	Hole-making	#39 Drill Bit	Create the holes for the pins on the Mustache

Post-Processing

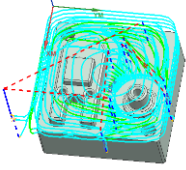
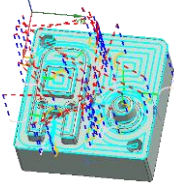
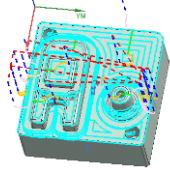
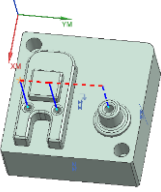
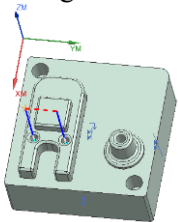
The $\frac{1}{8}$ in. diameter and $\frac{1}{2}$ in. long pin was press fit using the Arbor Press.

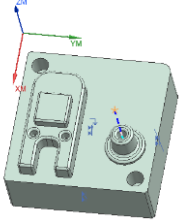
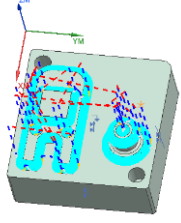
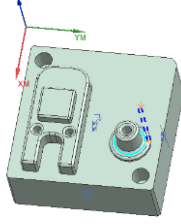
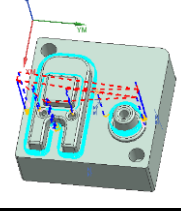
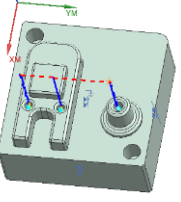
Table 3: Cavity mold for the Front and the Mustache CAM Processes & Operations

Process	Parameters	Tool	Discussion
3D Adaptive Milling 	Mill Rough	$\frac{3}{4}$ in Flat End Mill	Remove material
3D Adaptive Milling 	Mill Rough	$\frac{3}{8}$ in Flat End Mill	Remove more material around areas with curvature less than $\frac{3}{4}$ in
Floor wall 	Mill Semi-rough	$\frac{3}{8}$ in Flat End Mill	Mill out the inner fillets inside the Cavity of the Among Us Front
Cavity Mill 	Mill-Finish	$\frac{1}{4}$ in Tapered Flat End Mill	Create the draft around the Among Us Front Cavity
Cavity Mill 	Mill-Finish	$\frac{1}{8}$ in Ball End Mill	Mill out the Mustache Cavity and the $\frac{1}{8}$ in fillets inside the Among Us Front Cavity
Cavity Mill 	Mill-Finish	$\frac{1}{4}$ in Ball End Mill	Make the runners and the mill inside the visor
Wall Profiling 	Mill-Finish	$\frac{1}{8}$ in Ball End Mill	Make the pin the top of the Among Us Front

			
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Table 4: Core mold for the Back and the Hat CAM Processes & Operations

Process	Parameters	Tool	Discussion
3D Adaptive Milling 	Mill Rough	$\frac{3}{4}$ in Flat End Mill	Remove material
Cavity Mill 	Mill Semi-Rough	$\frac{1}{2}$ in Flat End Mill	Remove more material around the Among Us Back and Hat Core
Cavity Mill 	Mill Semi-Rough	$\frac{1}{4}$ in Flat End Mill	Remove material in areas that the $\frac{3}{4}$ in Flat End Mill could not get into
Spot Drilling 	Center Drilling	Center Drill	Center drill on areas that will be drilled out in the following operations
Drilling 	Hole-making	#31 Drill Bit	Create the holes for the pins in the Among Us Back
Drilling	Hole-making	#30 Drill Bit	Create the holes for the

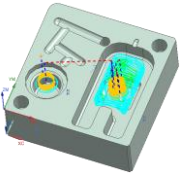
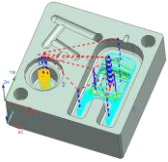
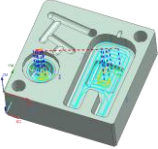
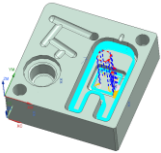
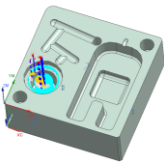
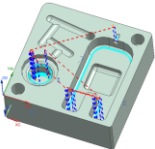
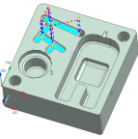
			pins on the Hat
Area Mill 	Mill-Finish	¼ in Ball End Mill	Finish operation to create a good surface finish on the Among Us Back and Hat
Wall Profiling 	Mill-Finish	⅛ in Ball End Mill	Mill out the ⅛ in fillets on the Hat
Cavity Mill 	Mill-Finish	⅛ in Tapered Flat End Mill	Create the draft around the Among Us Front
Wall Profiling 	Mill-Finish	⅛ in Tapered Flat End Mill	Create the draft inside the Among Us Back and inside the stepped pin

Post-Processing

The ⅛ in. diameter and 1 in. long pin was press fit using the Arbor Press.

Table 5: Cavity mold for the Back and the Hat CAM Processes & Operations

Process	Parameters	Tool	Discussion
3D Adaptive	Mill Rough	½ in Flat End	Remove material

Milling 		Mill	
3D Adaptive Milling 	Mill Rough	$\frac{3}{8}$ in Flat End Mill	Remove more material around areas with curvature less than $\frac{1}{2}$ in
Floor wall 	Mill Semi-rough	$\frac{3}{8}$ in Flat End Mill	Mill out the inner fillets inside the Cavity of the Among Us Back
Cavity Mill 	Mill-Finish	$\frac{1}{4}$ in Tapered Flat End Mill	Create the draft around the Among Us Back Cavity and the Hat Cavity
Cavity Mill 	Mill-Finish	$\frac{1}{4}$ in Ball End Mill	Make the runners and the mill inside the backpack
Cavity Mill 	Mill-Finish	$\frac{1}{8}$ in Ball End Mill	Mill out the Among Us and Hat Cavity and the $\frac{1}{8}$ in internal fillets.
Area Mill 	Mill-Finish	$\frac{1}{8}$ in Ball End Mill	Mill finish the pin at the top of the Among Us Back

Review of the CAM Processes and Machining Time

Core Mold for the Front and the Mustache (Machining Time: 59 mins)

During the machining of the core mold for the front and the mustache, an issue was identified with the top shutoff on the Among Us Front. The shutoff was too large, creating a larger hole diameter compared to the diameter of the Mustache. As a result, this caused a clearance fit instead of the intended press fit. To correct this, the shutoff was re-milled on the Haas Mini Mill 2, reducing the diameter by 0.005" smaller than the Mustache diameter. Additionally, the holes created for the pins were too small to allow for a proper press fit. To resolve this, a manual milling operation was performed using a reamer of 0.1245" to drill out the 0.125" holes, ensuring the necessary fit. These adjustments contributed to some additional machining time, particularly due to the reaming process, but ultimately resolved the fitment issues.

Cavity Mold for the Front and the Mustache (Machining Time: 56 mins)

The CAM operations for the cavity mold proceeded with minimal complications. However, during machining, the tapered end mill encountered momentary high loads at the corners, which caused loud noises. This was a brief issue but was noted as a potential cause of increased tool wear. While the machining went smoothly overall, some surfaces were left rough due to the machining process. To address this, Scotch Brite pads were used to hand-sand the affected surfaces, improving the surface finish and ensuring a smoother final product. This hand-finishing step contributed to additional time but was necessary to meet the required quality standards.

Core Mold for the Back and the Hat (Machining Time: 70 mins)

For the core mold of the back and the hat, there was an issue with the depth of the holes created to press-fit the pins. The initial hole depth was insufficient to allow the 1-inch pin to fit flush with the top of the Hat Cavity. To fix this, a #30 drill bit was used on the manual mill to drill the holes deeper. Additionally, the holes were still too small to achieve a press fit, requiring another manual milling operation with a 0.1245" reamer to enlarge the holes to the correct 0.125" diameter.

This additional operation required extra time and effort, particularly in drilling and reaming the holes manually, which prolonged the overall machining time. However, these steps were necessary to ensure a secure and accurate press fit for the pins.

Cavity Mold for the Back and the Hat (Machining Time: 68 mins)

The CAM operations for the cavity mold of the back and the hat also proceeded largely without issues. Similar to the front and mustache cavity mold, the tapered end mill encountered brief high loads at the corners, resulting in loud noises. While this was a temporary issue, it was noted that the tool had to work harder in these areas, which could increase machining time or tool wear. As with the front and mustache cavity mold, Scotch Brite pads were used to hand-sand rough surfaces, contributing additional time to the process but ensuring a smooth and uniform finish across the cavity mold.

Conclusion

Overall, the CAM processes for both the core and cavity molds proceeded relatively smoothly, though several issues arose that impacted machining time. Adjustments such as re-milling the shutoff, drilling

deeper holes, and reaming the holes to the correct diameter were necessary and added time to the machining process. Additionally, hand-finishing steps using Scotch Brite pads ensured the molds had the desired surface finish, though this added to the overall machining time. Despite these challenges, the final results met the necessary design and fit specifications. The additional time spent in troubleshooting and finishing was essential to ensure the quality and functionality of the molds.

Review of Molding Process

Table 6: Molding Parameters Used to Make the Among Us Back and the Hat

To get to the parameters, some preliminary parameters were arbitrarily chosen and run on the mold. The preliminary values:

- Injection time: 4.5 sec
- Injection Pressure: 3000 psi
- Injection Temperature: 450 F

Parameter	Value
Melting temperature (F)	500
Injection time (s)	4.2
Cooling time (s)	15
Injection Pressure (psi)	3750

These parameters led to short shots and in order to fix that, the team increased pressure and temperature gradually. The pressure was increased to 3750 psi where short shots were no longer observed on the Among Us Back but the Hat would not fill all the way. Temperature was increased to 500 F which resulted in flashes but filled both the Back and the Hat. The injection time was then decreased from 4.5 sec to 4.2 sec to avoid these flashes.

Table 7: Molding Parameters Used to Make the Among Us Front and the Mustache

To get to the parameters, some preliminary parameters were arbitrarily chosen and run on the mold. The preliminary values:

- Injection time: 5 sec
- Injection Pressure: 3250 psi
- Injection Temperature: 475 F

Parameter	Value
Melting temperature (F)	487
Injection time (s)	7.3
Cooling time (s)	30
Injection Pressure (psi)	2750

These parameters led to short shots in the Among Us Front while the Mustache was getting completely filled. To fix that, we increased the pressure and temperature gradually. The pressure was increased to 3500 psi where short shots were no longer observed on the Among Us Front and Mustache but flashes were observed since the ideal injection time was about 4.18 sec (very difficult to get). Temperature was increased to 487 F and pressure reduced to 2750 psi to allow a longer injection time (7.3 sec, easier to hit, and even if the exact time was not hit, defects would not be observed) without resulting in flash for both the Front and the Hat.



Figure 8: Fully assembled parts

Following the whole injection molding process, the part assembled well especially in areas where inference fits were required at the Hat, Mustache and between the Among Us Front and Back Bodies.

However, there is a small gap between the Front and Back Bodies at the legs which suggests that adding pins and pin holes would have been beneficial at those areas.

Some difficulties encountered during the injection process:

- Removing the Hat from the mold
- Collecting exact injection time data
- Flash from the vents
- More colored pellets than clear pellets caused variations in the ideal molding parameters

Metrology

As part of our metrology study, our team selected four key dimensions to measure, corresponding to the four different parts in our design. Each dimension was chosen based on ease of measurement, ensuring that fillets, drafted walls, or complex geometry would not interfere with accurate readings, and on areas where we wanted to check for potential shrinkage during the injection molding process.

For the back half of the Among Us character, we measured the height of the backpack, as it is the only prominent feature on this part. On the front half, we measured the depth of the body, ensuring consistency in the overall character shape. For the hat, we measured its overall height to ensure it fits correctly onto the character without interference. For the mustache, we measured the diameter of its pin, as it is crucial for achieving a secure snap fit into the character's face and plays a key role in the final assembly. The pin must maintain a proper interference fit with its corresponding hole, ensuring stability while allowing for manufacturability. For the specified dimension and tolerance refer to *Table 9*.

Screenshots from our CAD models illustrating each measured dimension are shown below:

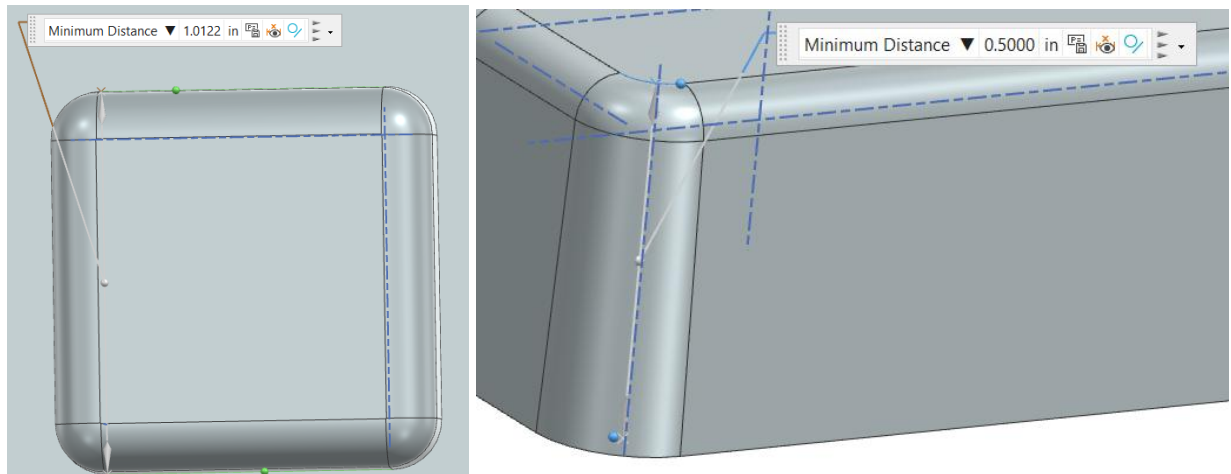


Figure 9: Backpack and Body Depth specified dimension in CAD

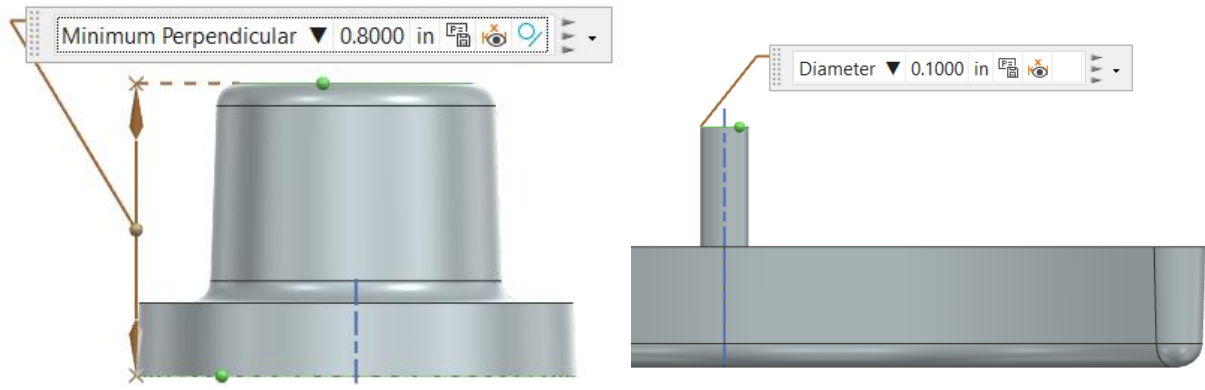


Figure 10: Hat height and Mustache pin specified dimension in CAD

Table 8: Production Data for All Parts

Part Number	Backpack Height (in)	Front Body Depth (in)	Hat Height (in)	Mustache Pin Diameter (in)
1	1.0065	0.4915	0.774	0.095
2	1.0045	0.4925	0.779	0.096
3	1.002	0.4905	0.78	0.0945
4	1.0025	0.488	0.775	0.095
5	1.0035	0.4875	0.779	0.0975
6	1.0005	0.4875	0.7725	0.096
7	1.0015	0.4895	0.771	0.099
8	1.0035	0.485	0.787	0.095
9	1.002	0.49	0.7775	0.0945
10	0.9995	0.491	0.7775	0.0945
11	1.001	0.4925	0.779	0.0965
12	1.0045	0.4885	0.7755	0.094

13	1.0025	0.4895	0.777	0.0955
14	1.0035	0.489	0.77	0.097
15	1.006	0.486	0.774	0.095
16	1.002	0.489	0.7725	0.096
17	1.006	0.484	0.774	0.0965
18	1.004	0.495	0.778	0.097
19	1.0045	0.486	0.7695	0.0975
20	1.0075	0.491	0.7785	0.096
Average	1.0034	0.4892	0.7760	0.0959
Standard Deviation	0.00211	0.00274	0.00411	0.00127

Table 9: Specified Dimensions For Measured Features

Feature	Specified Dimension	Tolerance ($\pm$ in)
Backpack Height	1.0122	0.01
Front Body Depth	0.5000	0.01
Hat Height	0.8000	0.01
Mustache Pin Diameter	0.1000	0.005

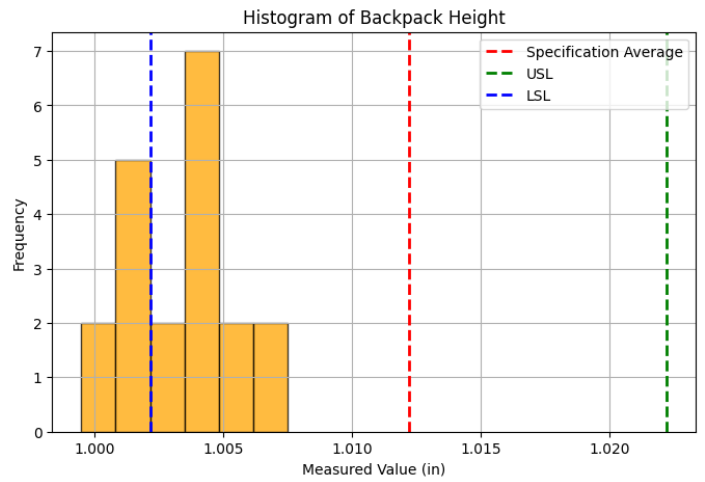
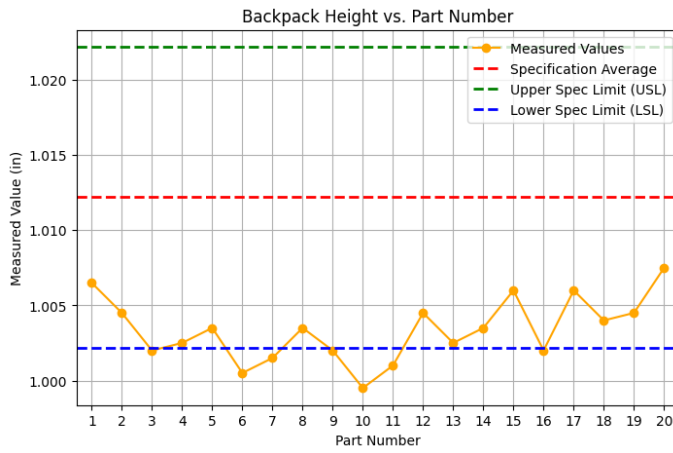


Figure 11: Backpack Height: Measured values vs. specification limits & histogram distribution

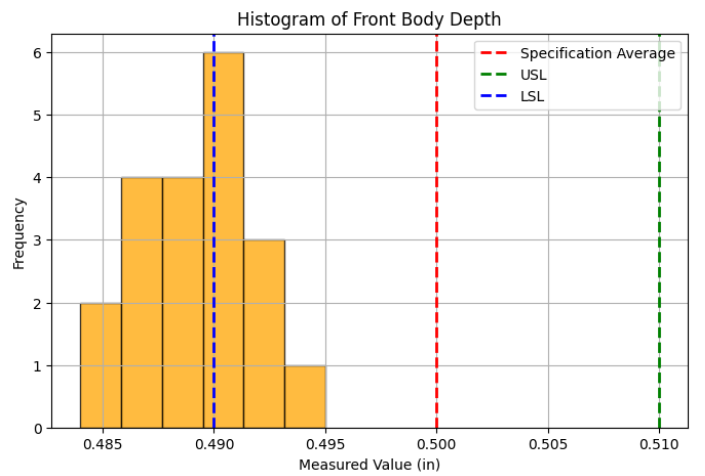
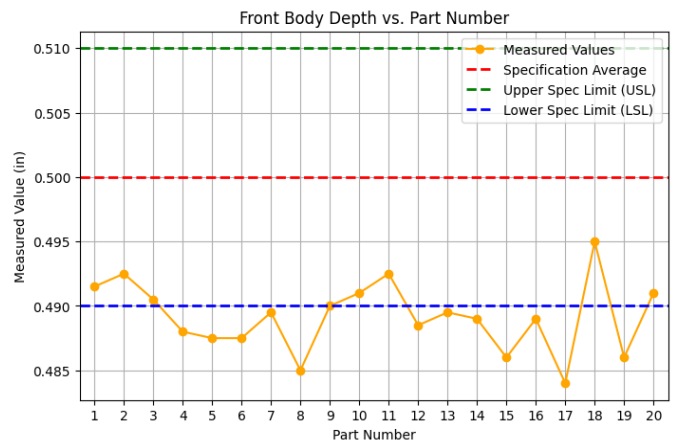


Figure 12 Front Body Depth: Measured values vs. specification limits & histogram distribution.

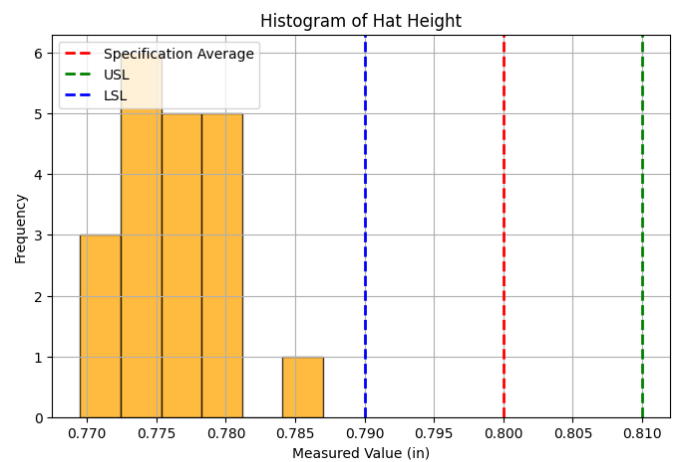
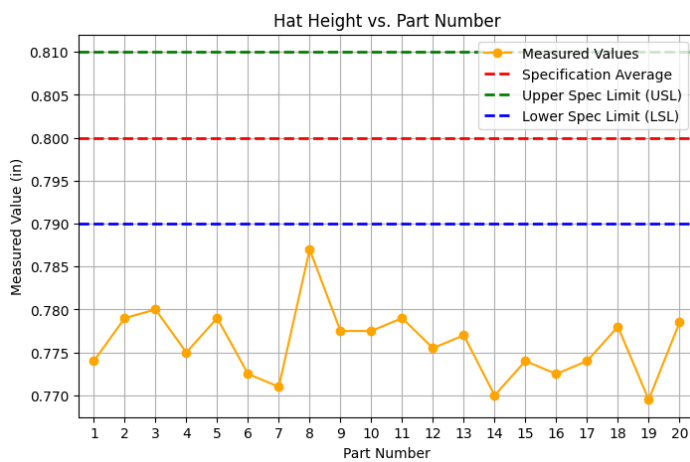


Figure 13: Hat Height: Measured values vs. specification limits & histogram distribution.

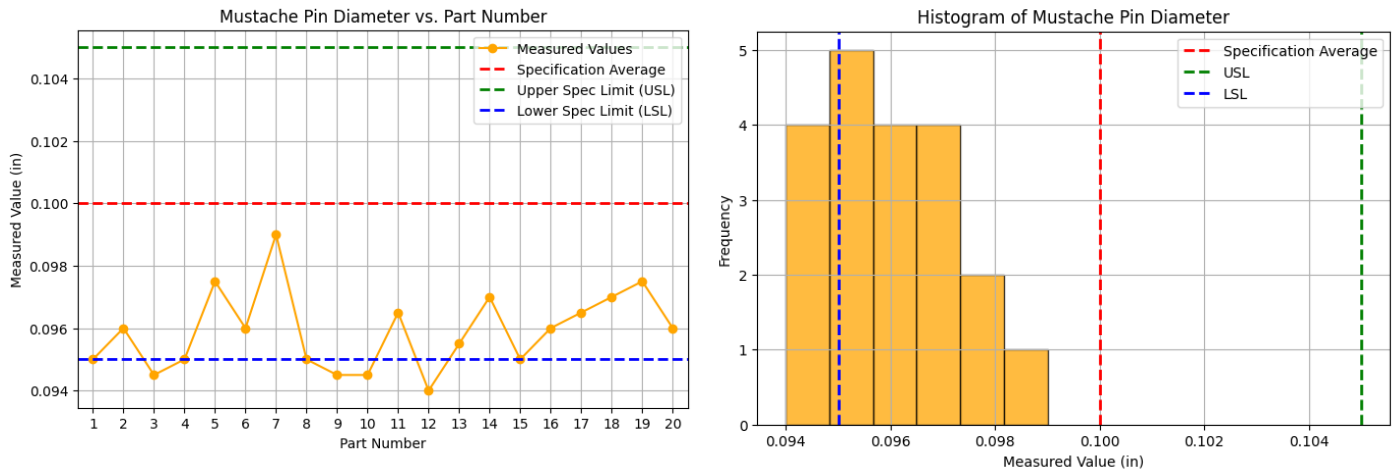


Figure 14: Mustache Pin: Measured values vs. specification limits & histogram distribution.

Table 10: Deviation of Part Measurements from Specification

Feature	Specified Dimension (in)	Measured Average (in)	Deviation (%)
Backpack Height	1.0122	1.0034	-0.88
Front Body Depth	0.5000	0.4892	-2.16
Hat Height	0.8000	0.7760	-4.1
Mustache Pin Diameter	0.1000	0.0959	-3.0

The molded parts show consistent deviation below the specified dimensions. Some of this variation is likely due to the extraction process, as a scraper had to be pushed underneath the hat while it was still hot, resulting in damage to the bottom surface.

These deviations align with the 1.0-2.5% expected shrinkage range for polypropylene, suggesting material contraction during cooling as the primary cause. However, all histograms show a normal distribution but are shifted left, meaning the measured values tend to be closer to the lower specification limit (LSL). This consistent shift suggests that shrinkage was more uniform rather than random, reinforcing that material behavior played a major role in dimensional reductions.

Aside from shrinkage, errors could have occurred during the mold manufacturing process. When machining the molds on the CNC, we may have unintentionally left excess material, which could have altered the final cavity dimensions. Additionally, variations in cooling rates across different parts of the mold may have led to further inconsistencies. The small sample size (n=20) limits our ability to make definitive statistical conclusions, but the consistent shrinkage trend suggests that the process is stable. Further data collection and process refinement would be necessary to confirm whether these deviations can be reduced through mold adjustments or changes in cooling conditions.

Conclusions & Suggested Modifications

This project provided valuable hands-on experience in the design, manufacturing, and testing of an injection-molded product. Our team successfully designed and fabricated molds to produce a customizable Plasticized Pretender with two attachable accessories. Throughout the process, we gained insight into the complexities of mold design, CNC machining, and injection molding, refining our understanding of material behavior, tooling, and process optimization.

Overall, our project was highly successful, as we were able to produce functional parts that assembled as intended with minimal defects. Our initial CAD models remained largely unchanged throughout manufacturing, and CAM operations were generated efficiently. CNC machining of the molds went smoothly aside from minor chatter issues due to high load on the tapered end mill at corners, but these did not significantly impact the mold quality.

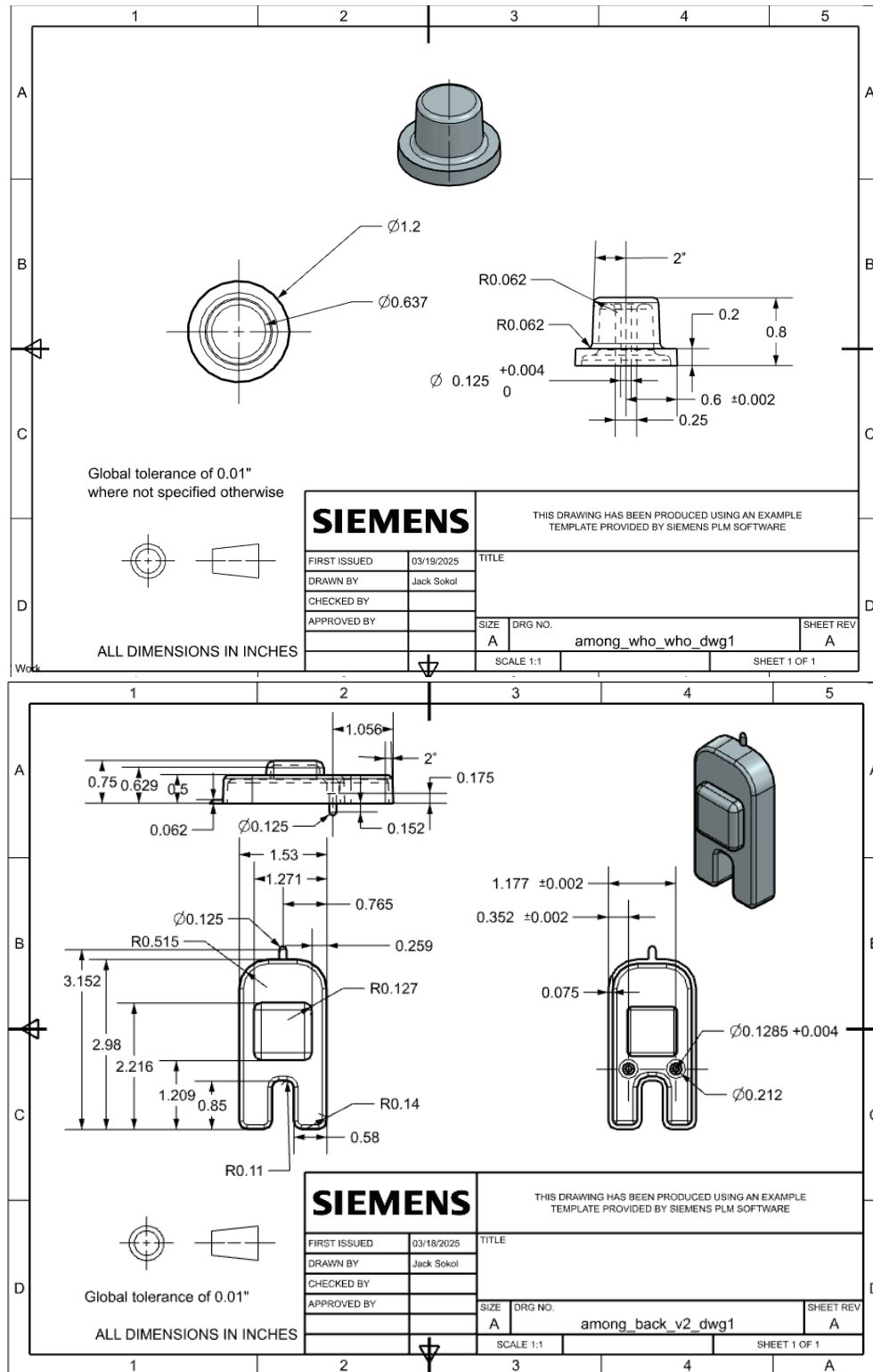
During injection molding, our biggest challenge was dialing in the correct process parameters. The top hat was particularly difficult to fill, as its runner was extended to balance the flow between the hat and the back half of the character. However, balancing runner distances is tricky, and our design was not perfect, this resulted in inconsistent fill quality, requiring adjustments to temperature, pressure, and injection time. Additionally, excessive plastic waste was generated from the sprues and runners. Since our parts were small, the volume of waste was greater than the material used for each part, which would be highly inefficient in a mass-production setting.

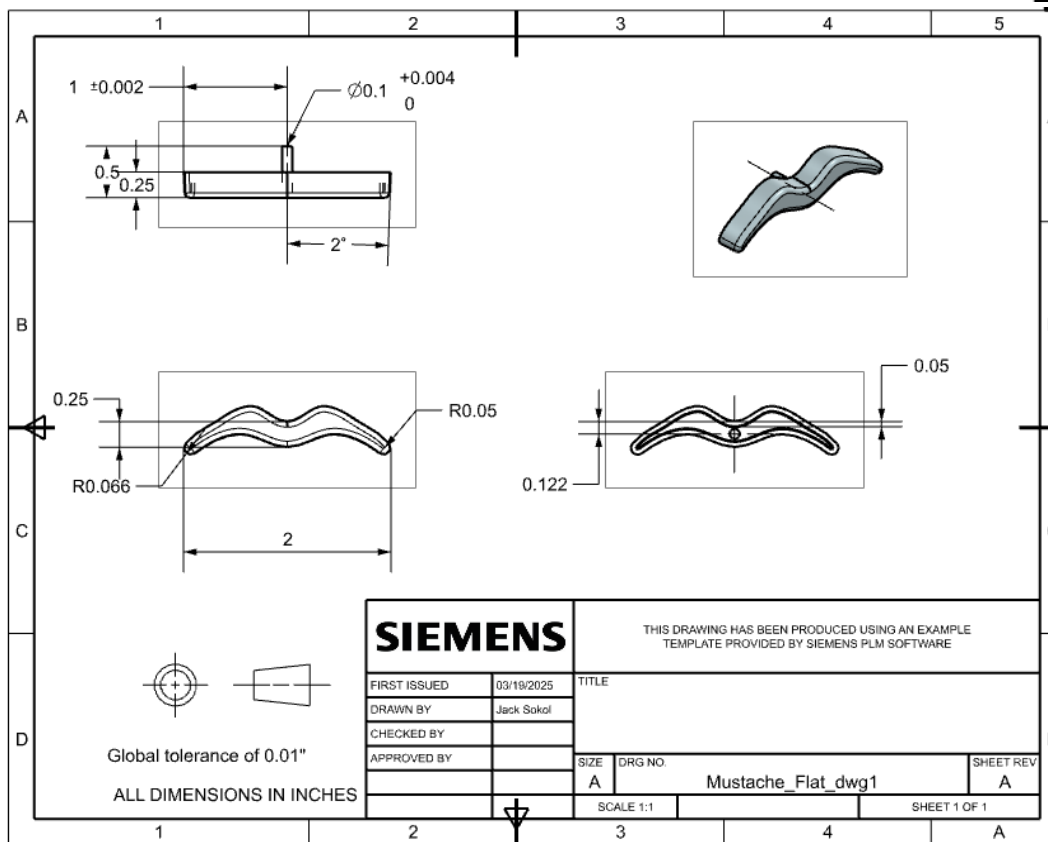
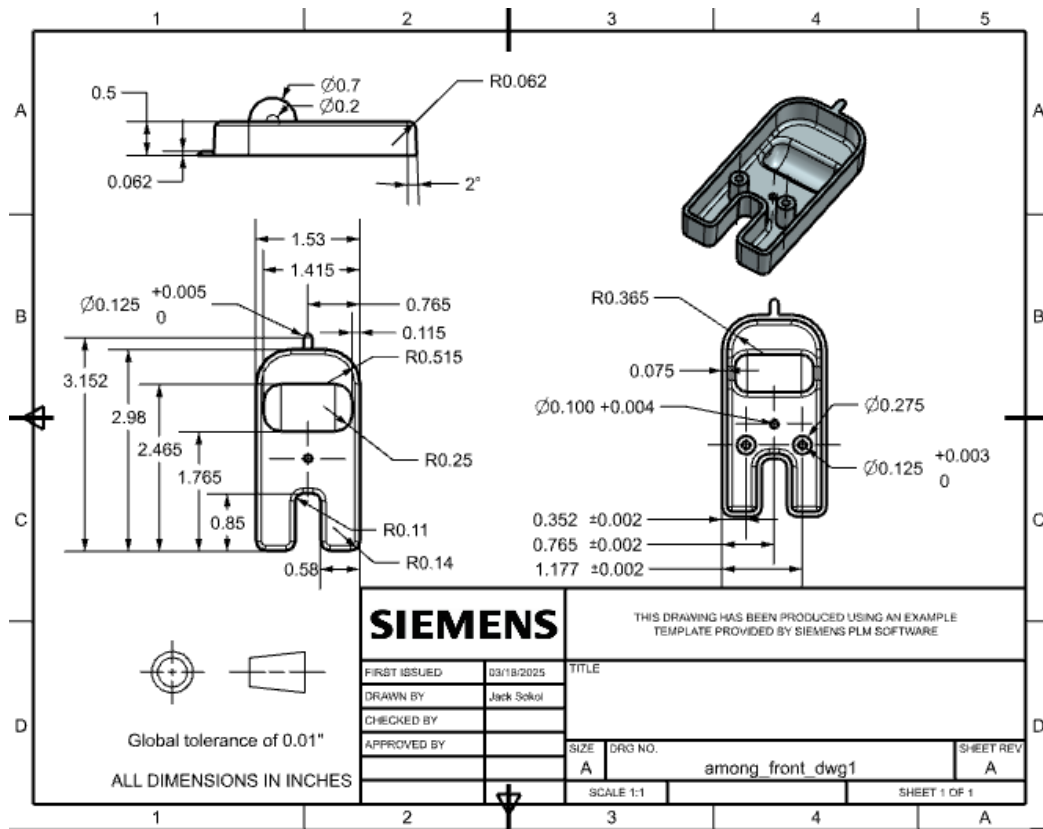
For future improvements, several mold modifications could help address these challenges:

- Optimizing runner design: Reducing runner length to the hat cavity could improve filling consistency without creating excessive back pressure.
- Adding more draft: At times, our parts would stick to the molds, making removal difficult. Increasing the draft angle would have helped facilitate smoother ejection by reducing friction between the part and mold walls. Additionally, ejector pins should have been incorporated to assist in pushing the part out of the mold consistently, preventing warping or damage during removal.
- Improving design for consistent tool use: Our mold required multiple different drill bit sizes, which made the drilling process more tedious and time-consuming. Standardizing hole sizes where possible would have reduced the number of tool changes, improving efficiency in the CNC machining phase and making the mold fabrication process more streamlined.

Despite these challenges, we are very satisfied with our final results. The experience gained in CAD modeling, CAM programming, CNC machining, and process tuning provided a deeper understanding of injection molding and part design. Moving forward, applying these lessons to future designs will allow for more efficient mold fabrication and process control, ultimately leading to better part quality and higher manufacturing efficiency.

Appendix: CAD Drawings of Parts & Molds





CAD Drawings of Mold Parts

